Explorer Otto

Álamo

VIEW IN BROWSER

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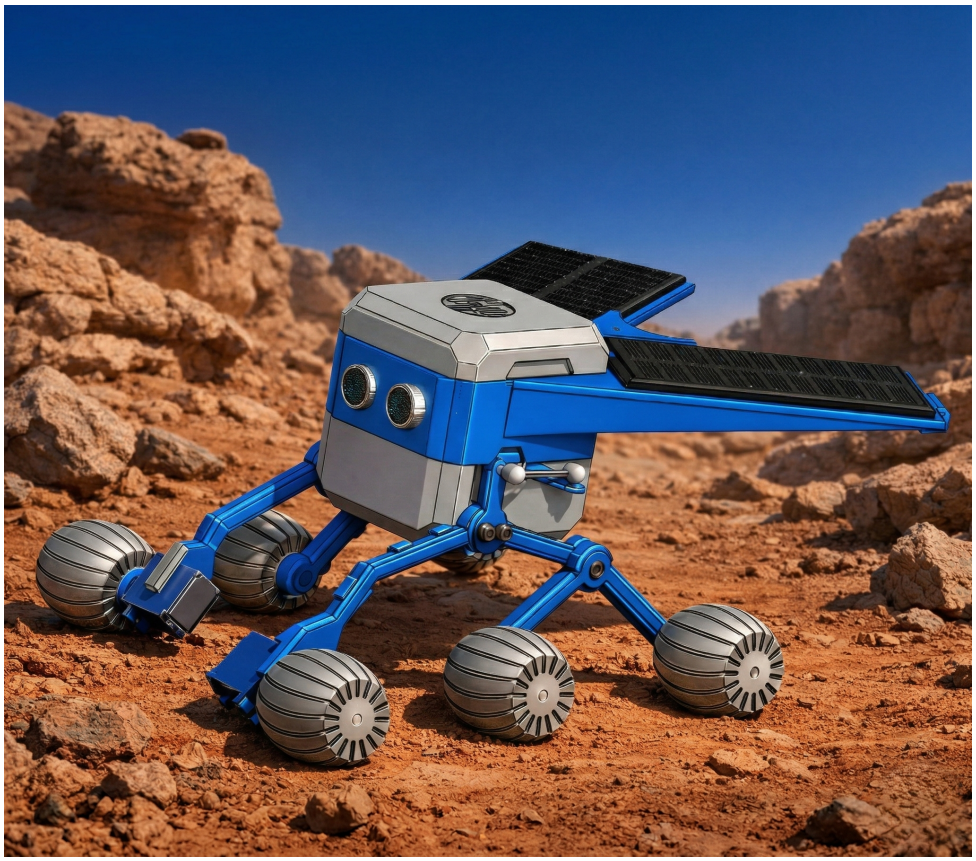
Summary

Otto DIY mod: I built the Explorer Otto with a rocker-bogie system and solar panels for off-road exploration.

Hobby & Makers > RC & Robotics

Tags: exploration otto robotics rover

Otto has been getting more and more curious. Beyond its versatility, our beloved robot has started to feel the urge to explore new terrains. So, I decided to give it a little push and create some modifications so it could embark on new adventures.



Introducing **Explorer Otto**, a complete modification that transforms the friendly base we all know into an off-road exploration vehicle.

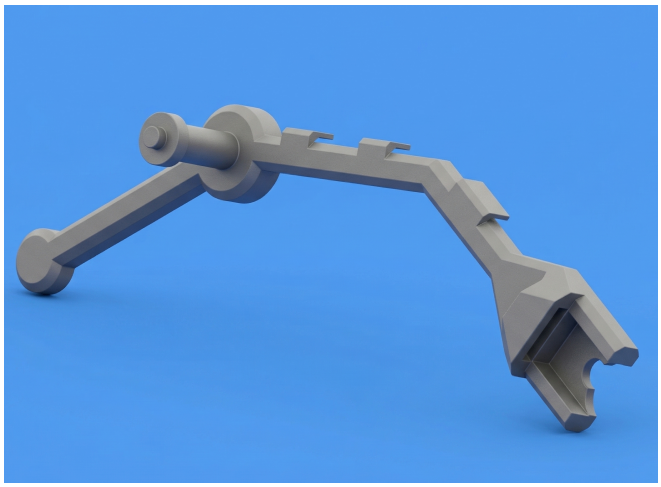
The main idea was to give Otto the ability to traverse complex obstacles. To achieve this, I developed and adapted:

- **Rocker-Bogie System:** I replaced the traditional legs with a mechanical suspension inspired by Martian rovers. This allows Otto to keep all wheels in contact with the ground while climbing rocks and uneven terrain.
- **Solar Panel Mount:** Every good space explorer needs energy, so I designed a dual rear mount to accommodate two solar panels (1W, 11x6cm), giving it that futuristic and functional “long-duration mission” look.

It was a challenge to design a fully 3D-printed rocker-bogie system. In my tests, it worked well, but you should follow a few **important tips**.

I recommend printing the system in **PETG** or **ABS**. All parts were designed to be printed without supports.

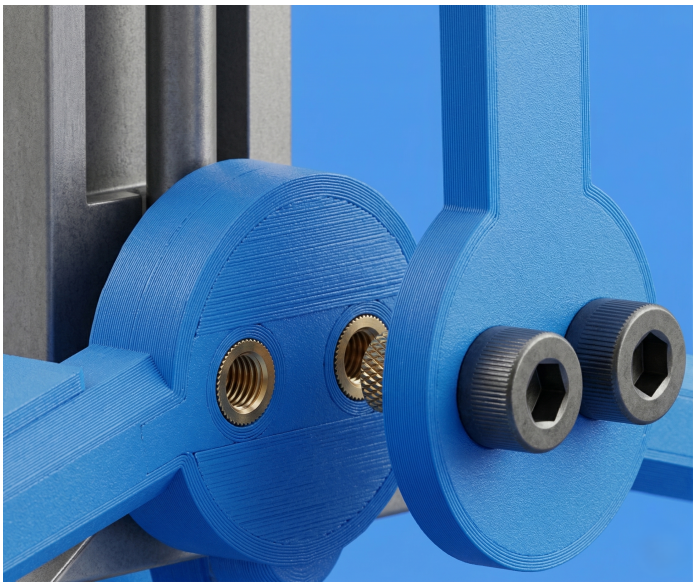
The **rocker** is the large component of the rocker-bogie system. I designed it to use the same motors from the original Otto kit, but in this version we need to place the motor on the outside of the robot. Therefore, I added small tabs that help guide the servo cables toward the holes located at the bottom of the bottom piece.

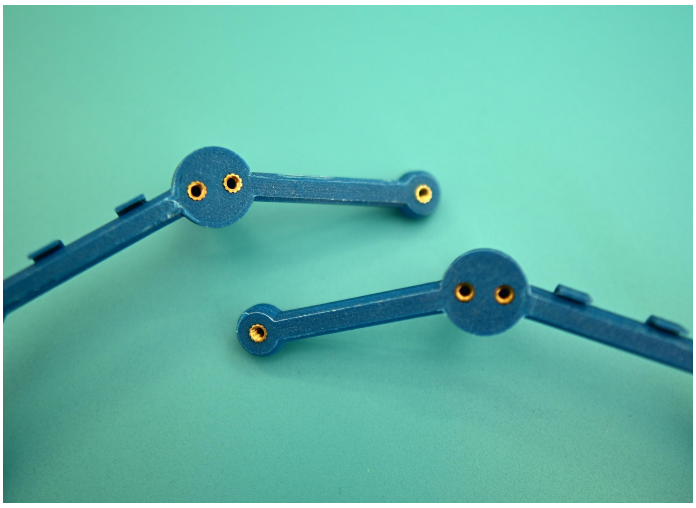


The remaining wheels attached to the bogie are not motorized; they rotate freely.

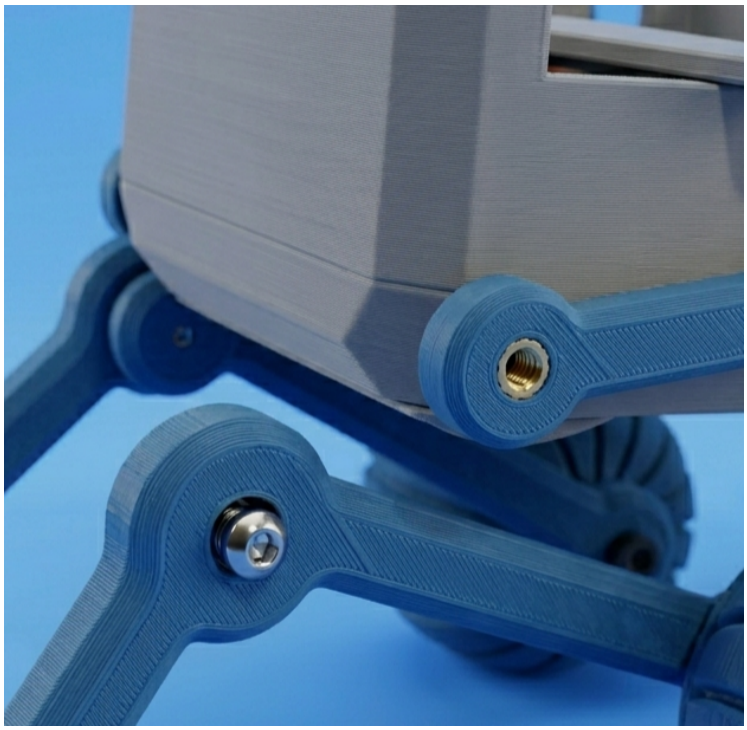
You must install heat inserts at the following points:

You must install 2× heat inserts on the Rocker–pivot joint. You may also try using self-locking screws here.



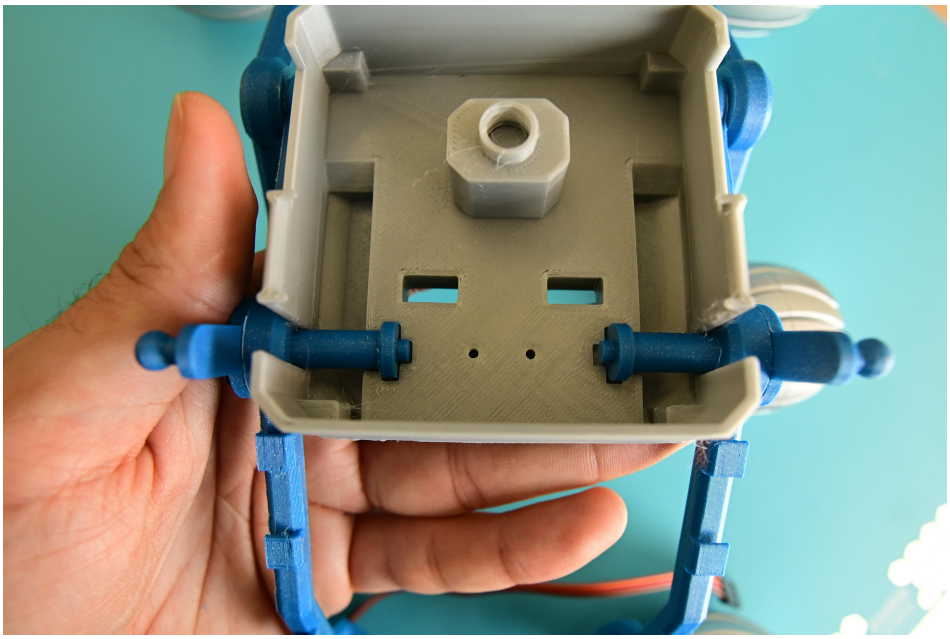


On each rocker, at the joint connection with the bogie, use 10 mm screws here. Tighten them fully — you should still be able to rotate the joint normally afterward.

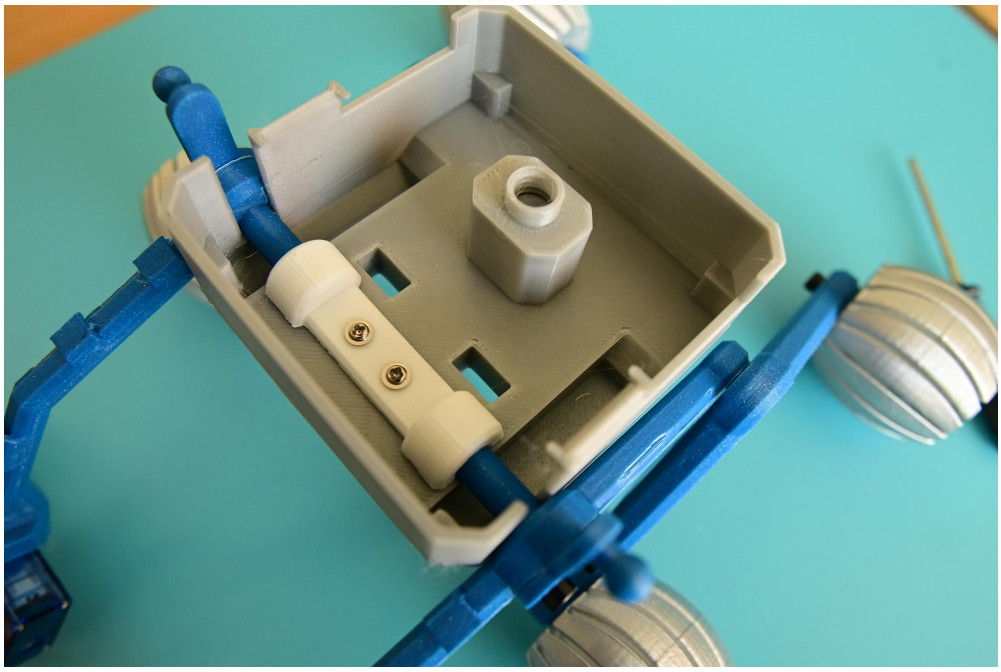


Now you can tighten the pivot with two M3 screws.

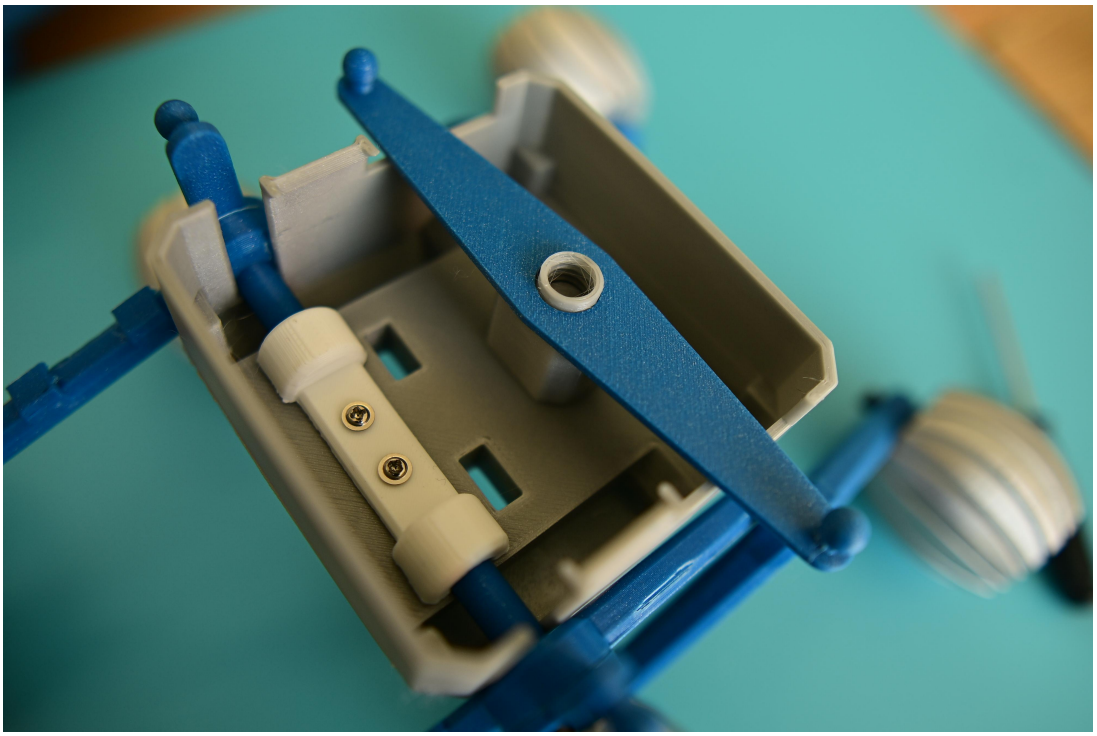
Inside the bottom piece is where the magic happens. You should position the two rockers in the cylindrical cavities and secure the locker on top with self-locking screws. Since this is where most of the friction occurs, it might be a good idea to apply some lubricant, even though I didn’t use any in my assembly.

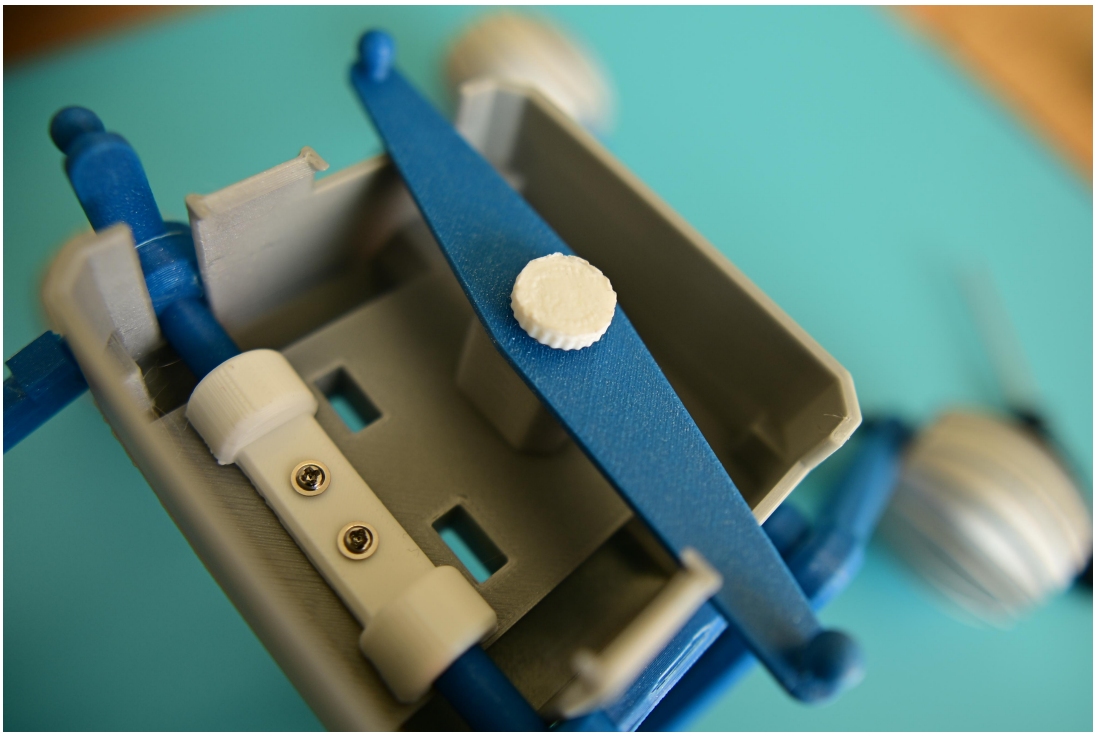


Now you can place the local_internal piece and tighten with self_locking screws



Place the 'differential' part onto the threaded cylinder and screw on the knob that locks the vertical movement.

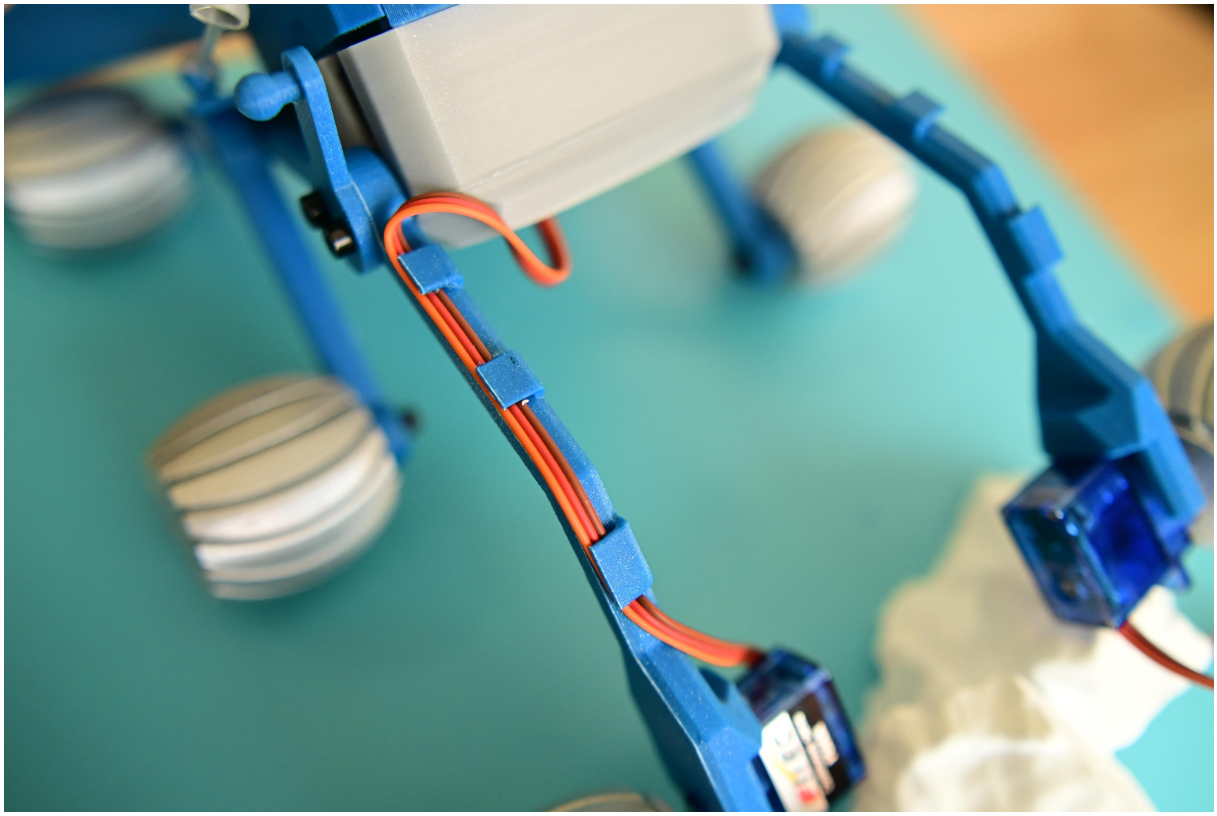




At this point, you can put the joints in place. I recommend printing them in PETG.

Align the modified version of the middle part and you will have the final assembly; note that the snap-fit is shorter because of the side opening for the 'differential' piece.

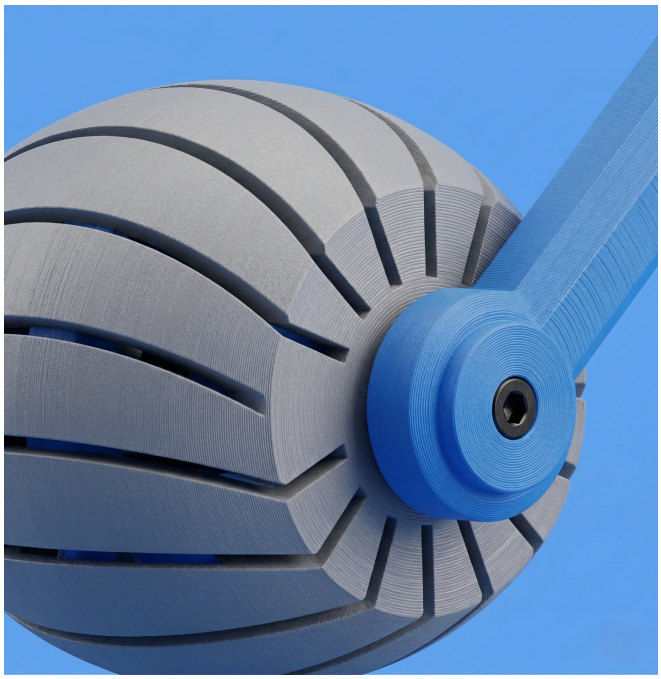
Use the small tabs on the rocker to organize the servo motor wires, then route the cable through the two holes on the bottom face of the 'bottom' part.



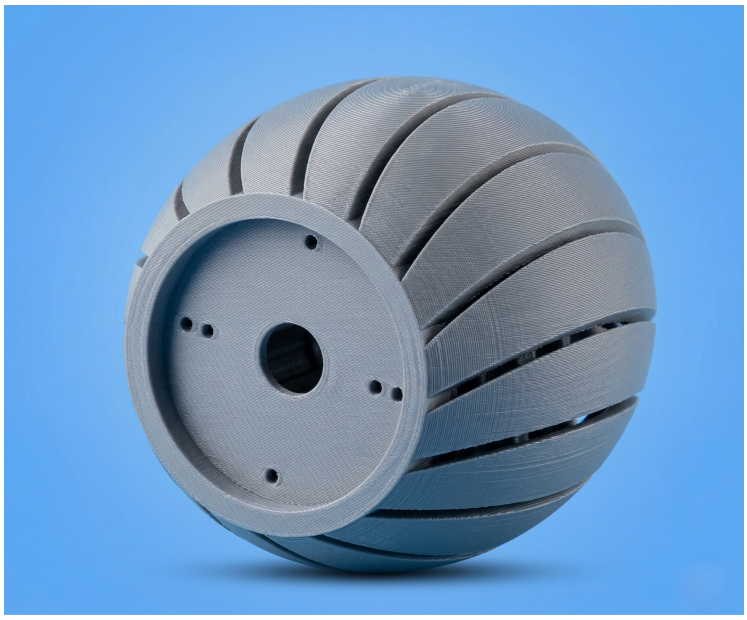
In my tests, for the free-spinning wheels, I installed a heat insert inside each wheel and used a 10 mm M3 screw. Just like in the other joint, you

can tighten it all the way — there’s still enough play to allow the wheel to rotate freely.

Another possible option is to place the heat insert in the bogie and use a longer screw that passes through the entire wheel.



For the motorized wheels, note that there is a separate file for them. You should use the mounting holes to attach the circular plastic servo adapter that comes with the motor.



You can experiment with different materials for these wheels. I tested them with 90A hardness TPU and was happy with the results. PETG also works, but keep in mind that the grip on rocks will be slightly compromised.

I designed the parts to place two 11x6cm solar pannels. I'm still waiting to receive the ones the i bought and i'll update the picutures soon :)

Have fun! I’d love to hear what you think of the assembly, and I’m looking forward to seeing your versions!

Model files

ottoexplorer_bottom.stl

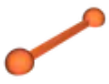
ottoexplorer_rocker.stl

ottoexplorer_bogie.stl

☐ symetrical, print 2 for assembly

ottoexplorer_middle.stl

rockerbogiesystem_diferential.stl



rockerbogiesystem_joint.stl

freespin_wheel.3mf

servo_motor_wheel.stl

solarpannel_lockpiece.stl

rockerbogie_pivot.stl

rockerbogie_locker_for_diferential.stl

locker_internal.stl

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